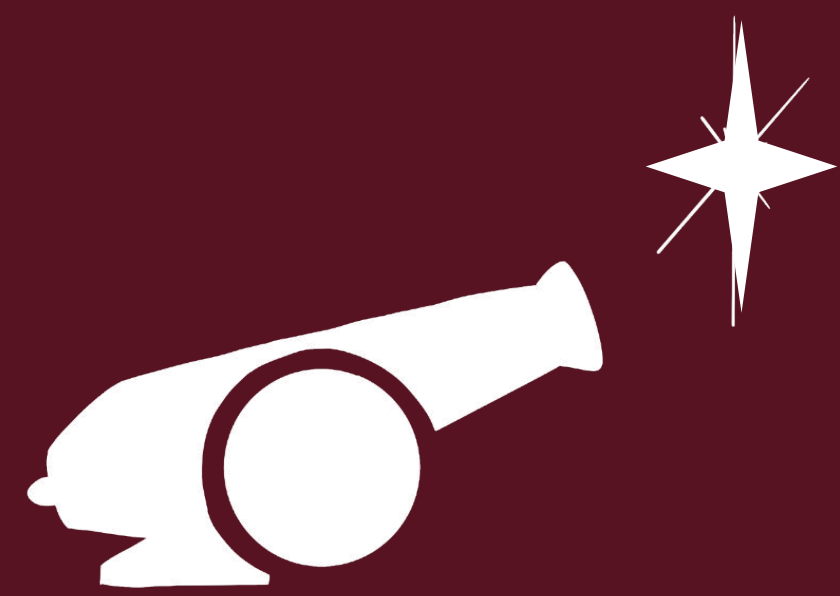




Stellar Label Estimate through use of Machine Learning

G. Leigh Harwell, Mason Beaty, Dr. Michael Hayden

Homer L. Dodge Department of Physics and Astronomy, Dodge Family College of Arts and Sciences

University of Oklahoma, Norman, OK, 73071

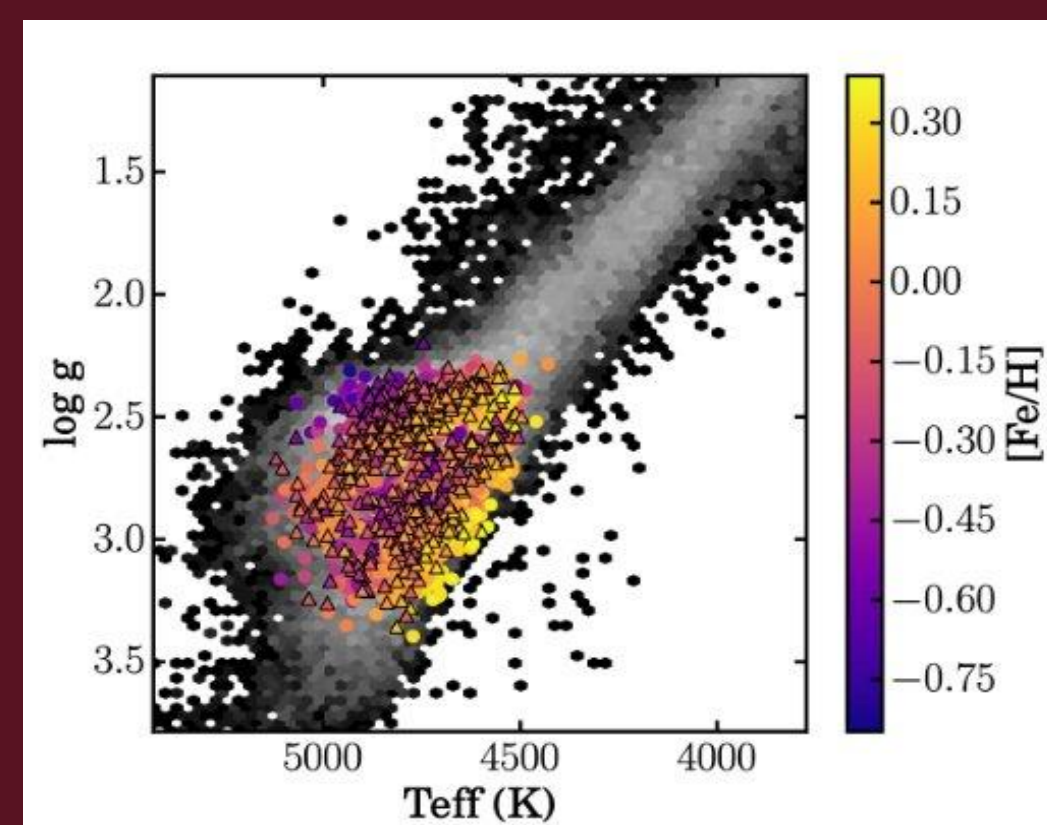


Introduction

- Red Giants are stars that have exhausted the hydrogen in the core and are extremely bright objects, making them easy to observe
- However, there are two kinds of Red Giants that are extremely similar

Red Giant Branch and Red Clump Stars

- Red Giant Stars (RGB)** Hydrogen burning shell, Low period spacing
- Red Clump Stars (RC)** Star that ignites helium fusion in their cores resulting in a “helium flash”, High period spacing



The 2 types of Red Giants on the HR-Diagram

Asteroseismology

Surface convection drives p-mode (sound wave) and g-mode (gravity wave) oscillations in the star. When these modes mix the time intervals between them are called period spacing. This can in turn inform us of what is happening in the interior of the stars.

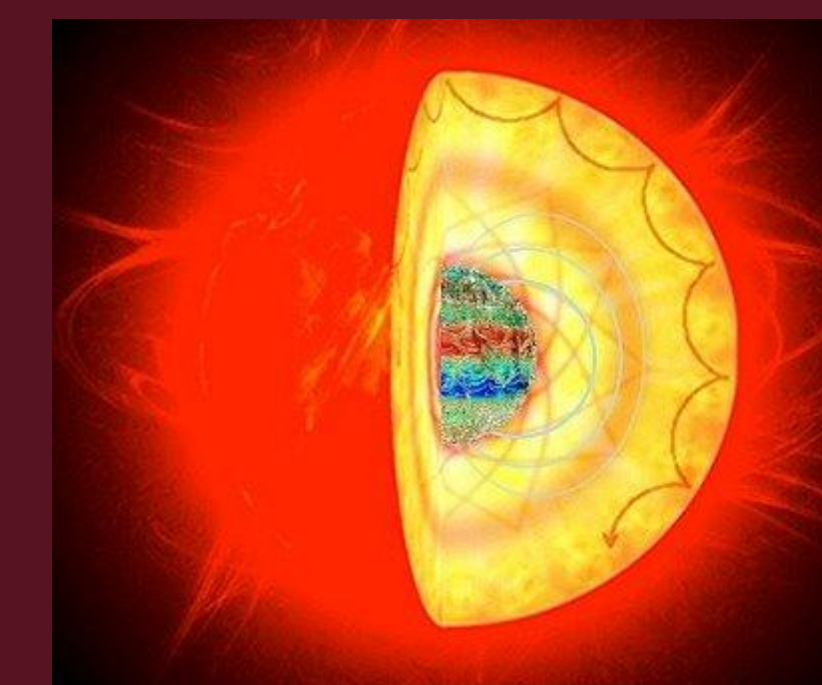


Figure 1: Image of how these modes act in the convection of the interior of a star.

Limitations

- Asteroseismology data provides deep insight but is expensive to gather and restricted to small celestial regions
- Other surveys like Gaia of the entire sky collecting spectral data for hundreds of thousands of stars
- By combining both types of data to build a predictive models that estimate stellar properties on a massive all-sky scale.

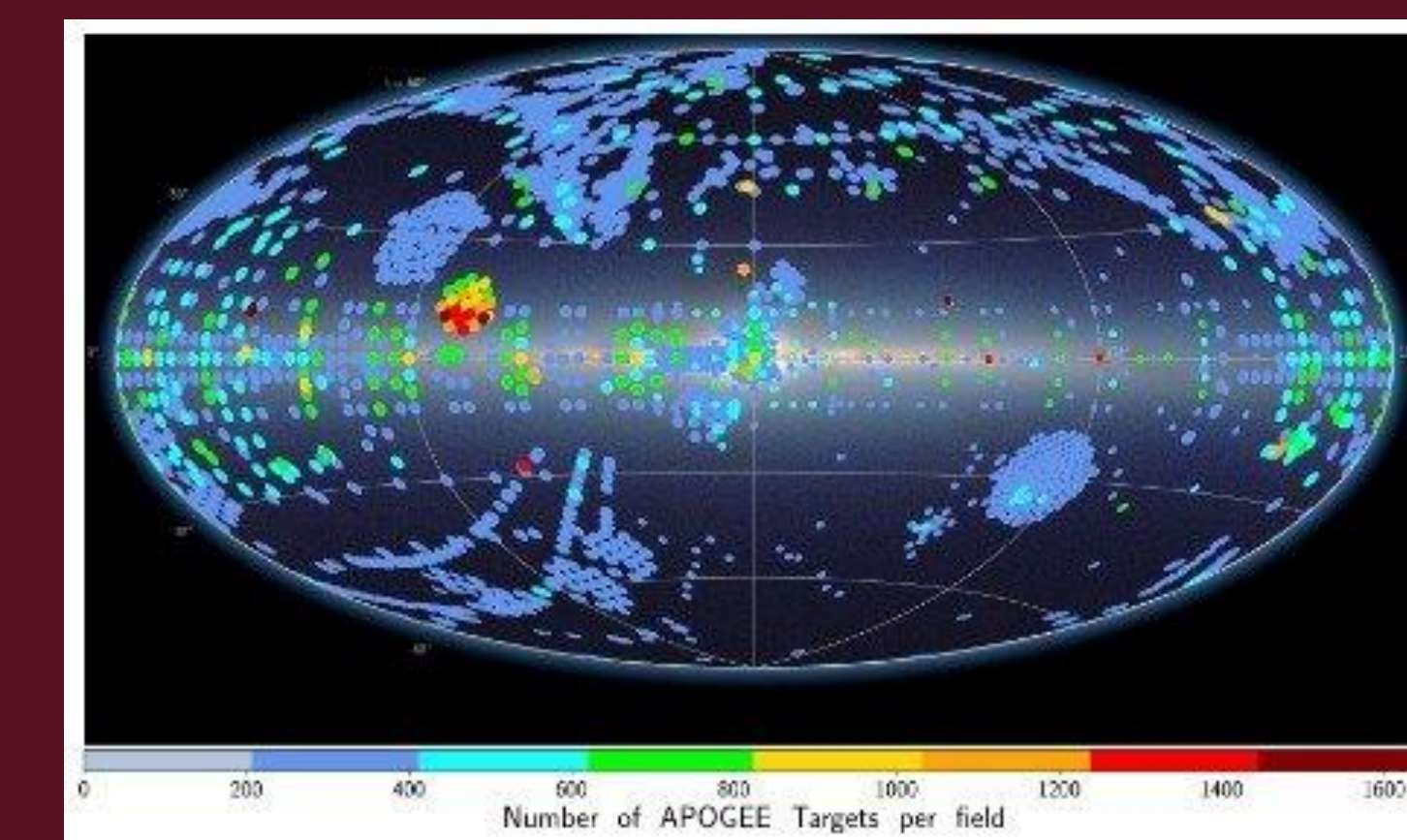
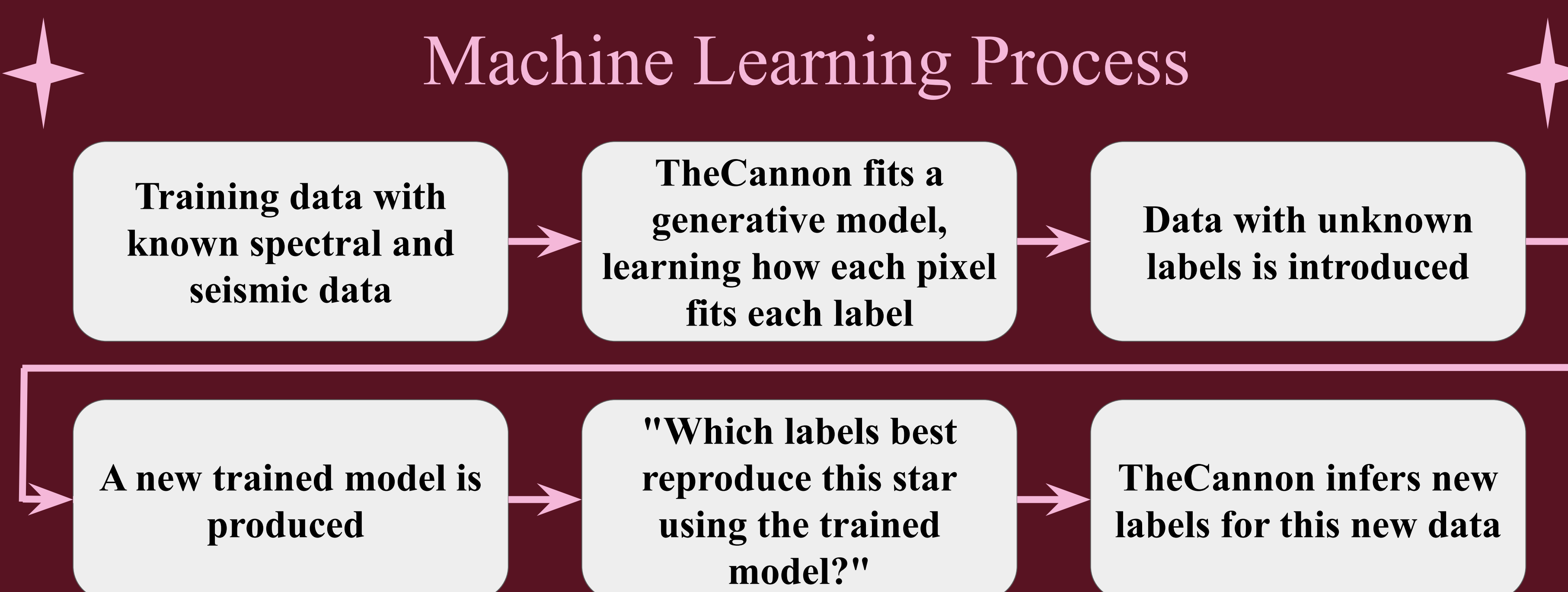


Figure 2: Observations of a full sky via APOGEE relative to where Kepler observes

TheCannon

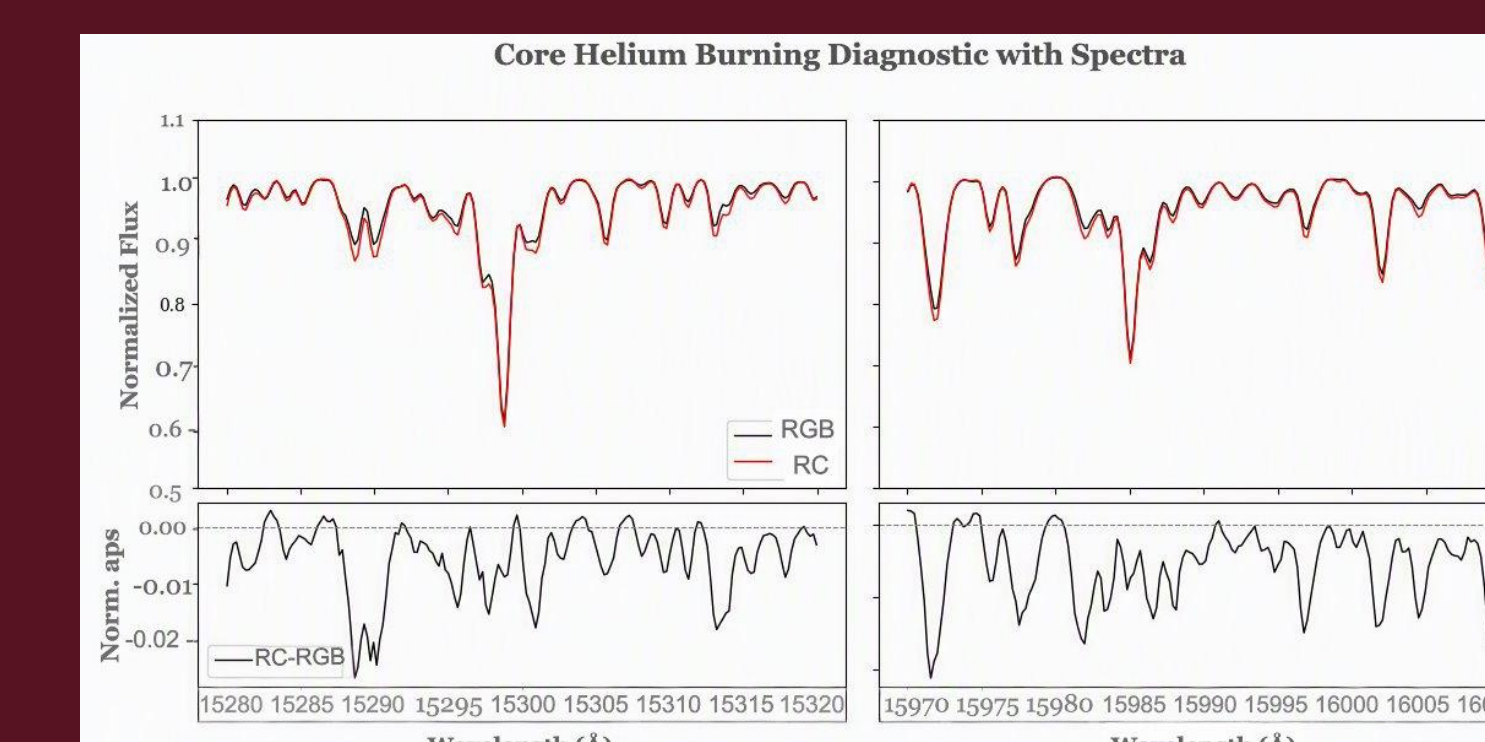
- TheCannon is a machine learning tool that can learn the relationship between a star's spectra and its known physical properties
- The training set is Kepler seismic data that is then applied to APOGEE spectra

Machine Learning Process



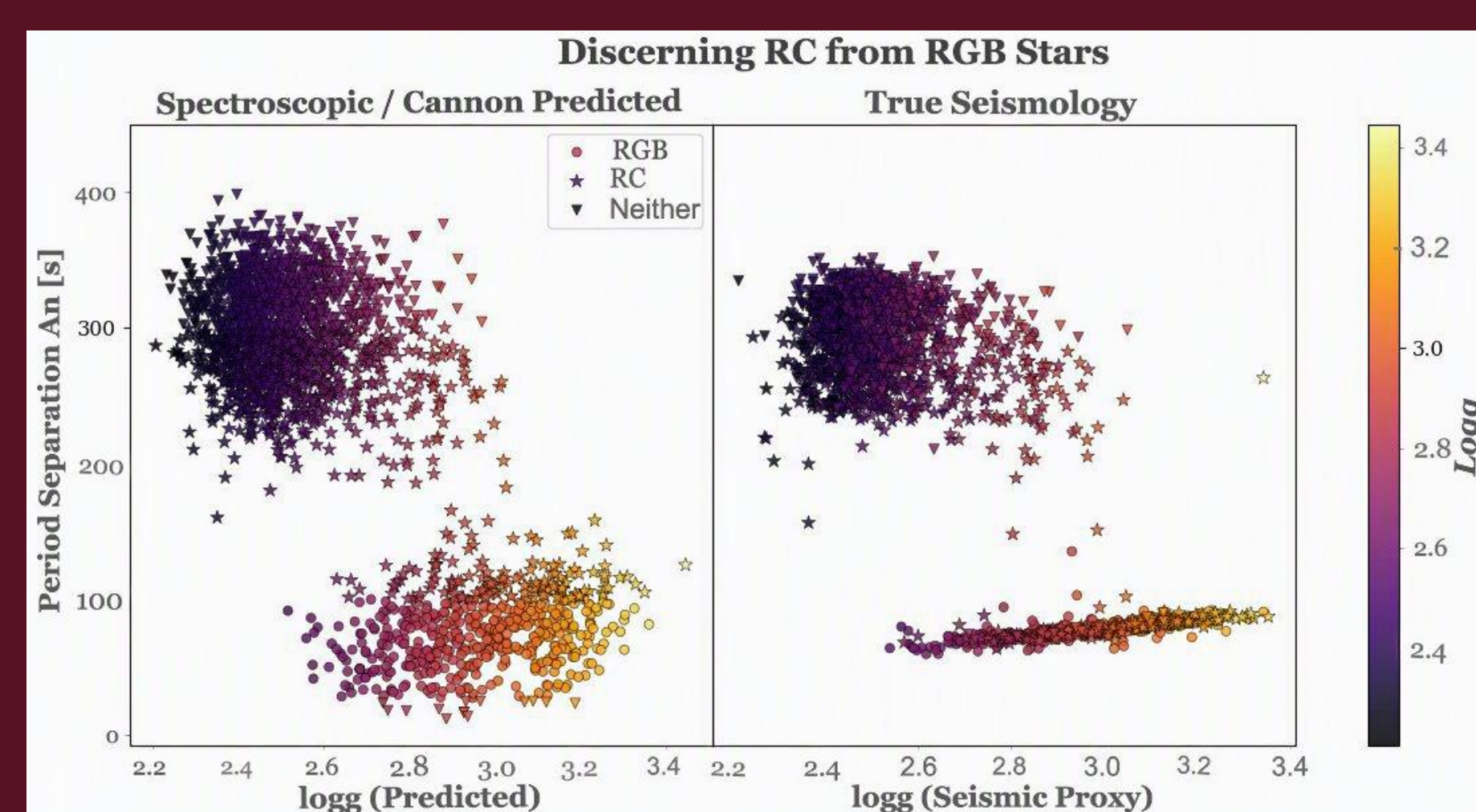
Why does this work?

The spectral region area shows that the differences between RGB and RC stars lie mostly around molecular CN and CO line features

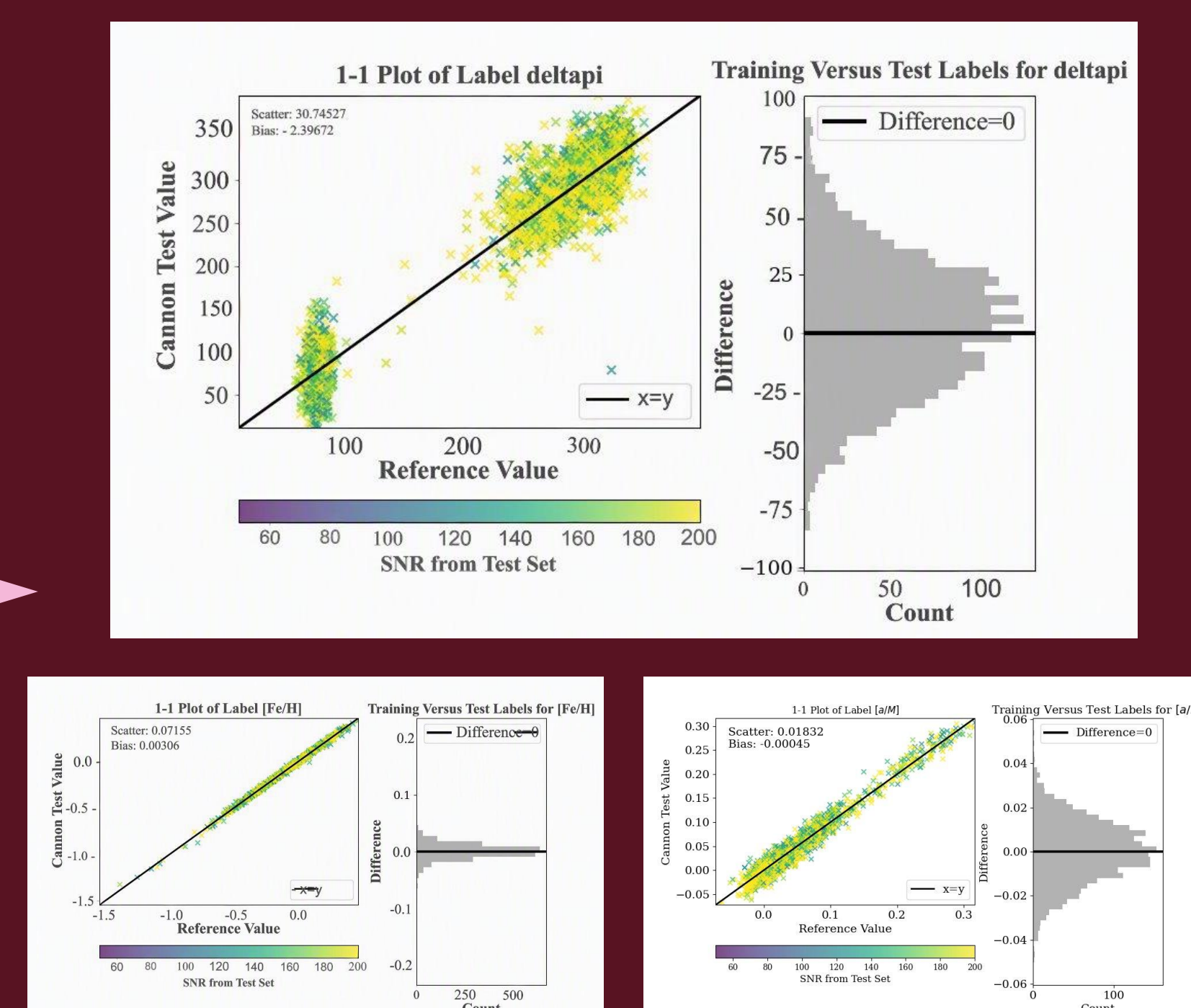


Results

- The separation between the RGB and RC stars being pretty evident and the histogram presenting like a gaussian
- Very small amount of uncertainty



Label Matching Results



Moving Forward

- Compare TheCannon's predicted labels with the data that the K2 telescope found
- Can the model predict these labels more accurately rather than what K2 observed
- Applying data to the larger APOGEE dr17 data set to apply the working model to a much larger data set

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